

Haidra Suleiman

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Portfolio: ai-personal-portfolio-flax.vercel.app | Full project portfolio: github.com/HaidraSuleiman

PROFESSIONAL SUMMARY

IT Engineer specializing in Data Science, Artificial Intelligence, and Web Development with experience developing data-driven solutions, AI applications, and modern web systems. Experienced in building scalable solutions using machine learning, automation, full-stack development, and modern software engineering practices. Passionate about solving complex problems through intelligent systems and technology.

PROFESSIONAL EXPERIENCE

IT Engineer – Data Science & AI | Welbond Armatures

2021 – Present

- Developing AI and data science applications using Python, machine learning, and data engineering principles
- Building data pipelines, retrieval systems, and automated workflows for data processing and analysis
- Applying statistical methods, data visualization, and predictive modelling to solve business problems
- Delivering end-to-end technical solutions integrating data processing, analytics, and AI technologies
- Developing and supporting web-based solutions to improve workflows, automation, and system accessibility
- Supporting technical infrastructure and implementing scalable software solutions across multiple domains

EDUCATION

Bachelor's Degree in Information Technology Engineering | Tishreen University

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, C++

Web Development: HTML, CSS, JavaScript, Next.js, React, FastAPI, Responsive Design, REST APIs

Data Science: Data Analysis, Data Processing, Statistical Analysis, Machine Learning

AI & LLM Technologies: Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Conversational AI

Tools & Technologies: LangChain, LangGraph, Vector Databases, ChromaDB, Pinecone, Supabase (pgvector), Azure AI Document Intelligence, Azure OpenAI, Docker, Gradio, Git, REST APIs, Pydantic, Streamlit

PROJECTS

Invoice Review — AI-Powered Invoice & Receipt Review System

End-to-end document intelligence app for a fictional EU facilities company; classifies, extracts, validates, and routes invoices/receipts for human review

- Built automatic invoice vs. receipt classification using Azure OpenAI structured output
- Combined Azure AI Document Intelligence extraction with an independent LLM pass, merged deterministically so Document Intelligence takes priority and the LLM only fills missing fields
- Implemented offline EU VAT format/checksum validation and totals reconciliation, plus a GL account suggestion engine against a fixed catalog with reviewer override
- Delivered a full review workflow (upload → process → review → history) with SQLite persistence and on-demand correction-email drafting
- Shipped a single-container Docker deployment (FastAPI + React SPA) to Azure Container Apps with Azure Files-backed storage

Tech Stack: Python, FastAPI, Pydantic v2, SQLAlchemy 2, SQLite, React, TypeScript, Tailwind CSS, Azure AI Document Intelligence, Azure OpenAI, Docker, Azure Container Apps

GitHub: [invoice-review](#) | **Live Demo:** [Document Review — Northstar Facilities](#)

Document Copilot — AI Chatbot for Document Q&A

Internal AI chatbot for a fictional investment research firm that lets analysts query a corpus of financial filings in plain English and get sourced, citable answers

- Designed a retrieval pipeline combining Supabase pgvector similarity search with Postgres full-text search for citable, sourced answers
- Built the backend with FastAPI and SQLAlchemy models, managing schema evolution with Alembic migrations against Supabase Postgres
- Integrated Supabase Auth for user access and OpenAI for LLM responses and embeddings generation
- Built a SEC EDGAR ingestion pipeline to populate the corpus with 10-K filings from major public companies (AAPL, MSFT, NVDA, AMZN, GOOGL)
- Deployed the full-stack app (FastAPI backend + React/Vite/TypeScript SPA) to Railway

Tech Stack: Python, FastAPI, React, Vite, TypeScript, Supabase (Postgres, pgvector, Auth), SQLAlchemy, Alembic, OpenAI, Railway

GitHub: [document-copilot-main](#) | **Live Demo:** [Document Copilot](#)

Adaptive RAG System — Corrective → Self → Adaptive Evolution — v1.0.0 Corrective RAG → v1.1.0 Self-RAG → v1.2.0 Adaptive RAG

- Designed and built an advanced multi-stage RAG system evolving through three architectural versions: Corrective RAG, Self-RAG, and Adaptive RAG
- Implemented intelligent query routing in v1.2.0 using a structured LLM router (Pydantic) to dynamically direct queries to a vector store or external web search
- Built hallucination detection and answer relevance grading loops (v1.1.0+) enabling self-correcting generation with iterative retry logic
- Engineered a multi-stage decision graph: query routing → retrieval → document grading → generation → self-evaluation → conditional correction
- Integrated ChromaDB vector store with LangGraph stateful workflows for production-style agentic pipeline orchestration

Tech Stack: Python, LangChain, LangGraph, OpenAI GPT-4.1-mini, ChromaDB, Pydantic, Prompt Engineering

LangChain Documentation Helper

Agent-based RAG assistant with automated documentation ingestion and semantic search

- Built a full end-to-end RAG assistant that crawls, chunks, embeds, and indexes the entire LangChain documentation into a Pinecone vector database for semantic search
- Implemented a LangChain Agent with a retrieval tool, enabling multi-step reasoning over documentation context before generating answers with source citations
- Engineered an asynchronous ingestion pipeline using Tavily Crawl and Extract to automate documentation collection, chunking, and OpenAI embeddings upload to Pinecone
- Delivered an interactive Streamlit chat interface with context-aware responses and source URL attribution for every answer

Tech Stack: Python, LangChain Agents, OpenAI GPT-4.1-mini, OpenAI Embeddings, Pinecone, Tavily, Streamlit, Asyncio

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LANGUAGES

English: Fluent | Arabic: Fluent